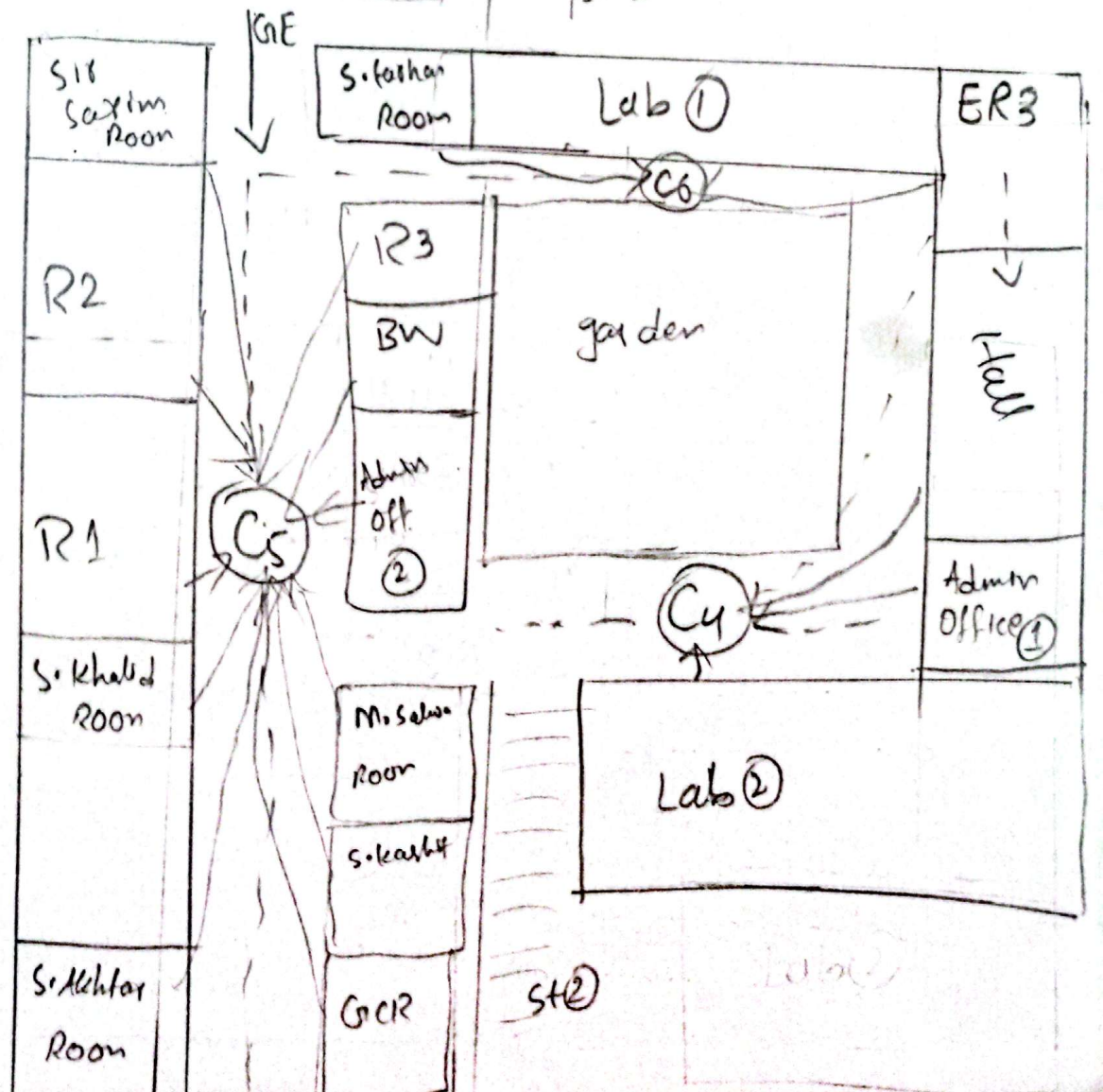
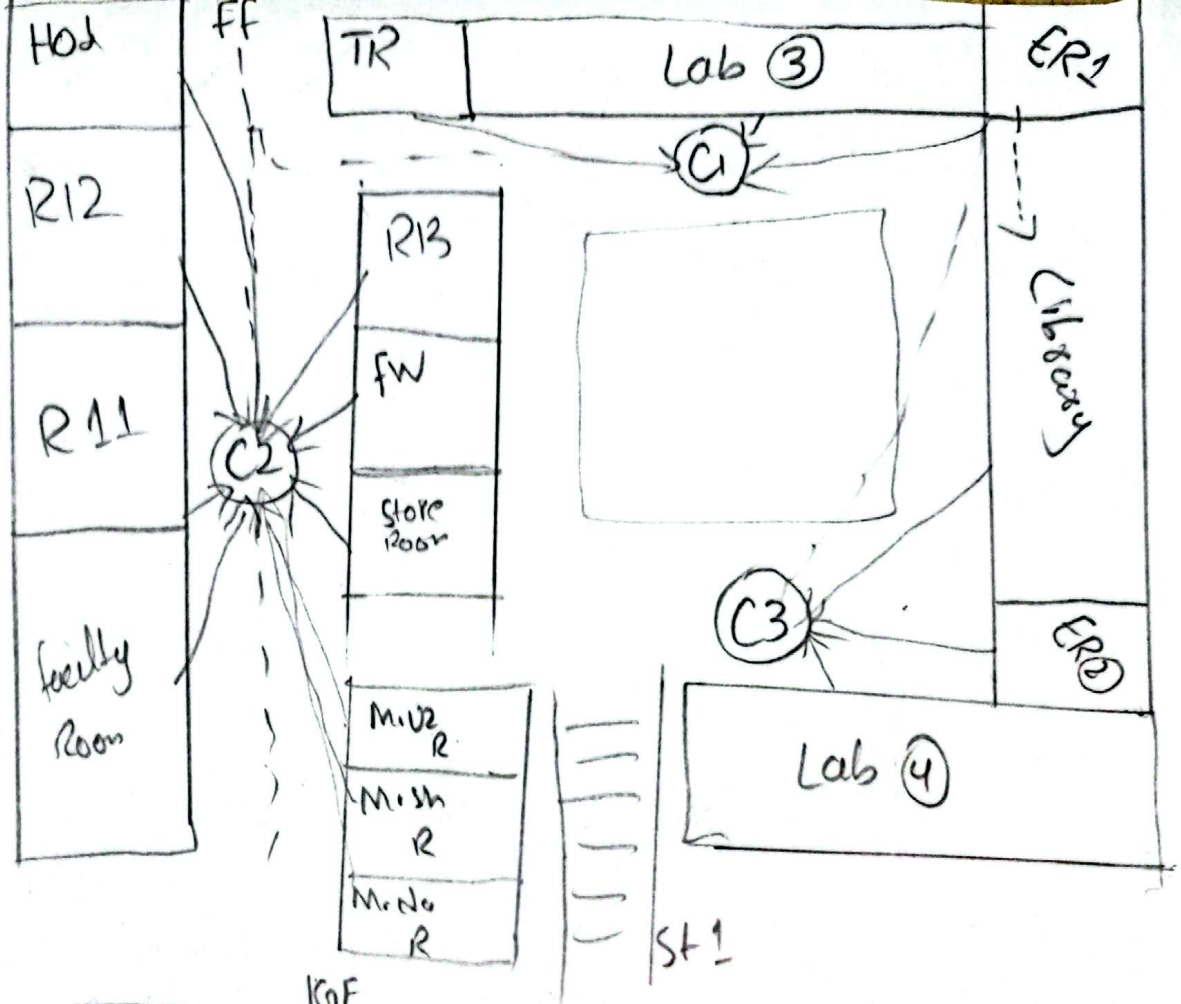


ground floor ← (Concept) → first floor



Transition Table:

FF = C₁, C₂, C₃

C₂ = Teacher's room, Lab ③, ER1, Library, C₂

ER1 → Library

Library → ER1, ER2

ER2 → C₃, Library, Lab 4, St 1

C₃ → Library, Lab 4, ER②, ER1, St 1

C₂ → HOD, JR13, R12, R11, Fw, Store Room, FR, MOR,

M-shR, M-NaR, C1

St 1 → St 2, C₃

GF → C₄, C₅, C₆

C₅ → S. Secdm R, R₃, R₂, R₁, BW, Adoff 2, S. K. L. R,

S. Alex R, C. C. R, S. K. L. R, M. L. R, C₆

St 2 → St 1, C₄

C₆ → S. L. R, Lab ①, ER③, Hall

C₄ → Lab 2, Adoff ①, Hall, St 2

HOD → C₂

R13 → C₂

R12 → C₂

R11 → C₂

Fw → C₂

SR → C₂

FR → C₂

MUR → C₂

M-shR → C₂

M-NaR → C₂

C1 → C₂

TR → C₁

Lab ③ → C₁

ER① → C₁

Ck → c1 ✓

C2 → C1

ER2 → C3

Lib3 → C3

Lab4 → C3

St1 → C3

lib → ER1

lib → ER2

St1 → C3

St2 → C4

S. Sam R → C5

R3 → C5

R2 → C5

R1 → C5

BW → C5

S. Kh R → C5

S. Akh → C5

GCR → C5

Sigleash → C5

Ms. Sul → C5

C5 → C6

St of ad R → C6

Lab① → C6

~~ER③~~ · ~~ER③~~ · ER③ → C6

Hall → C4, C6

Lab② → C4

adoff① → C4

adoff② → C5

Hall → adoff①

C4 → St2

Q formulate the problem?

Solve

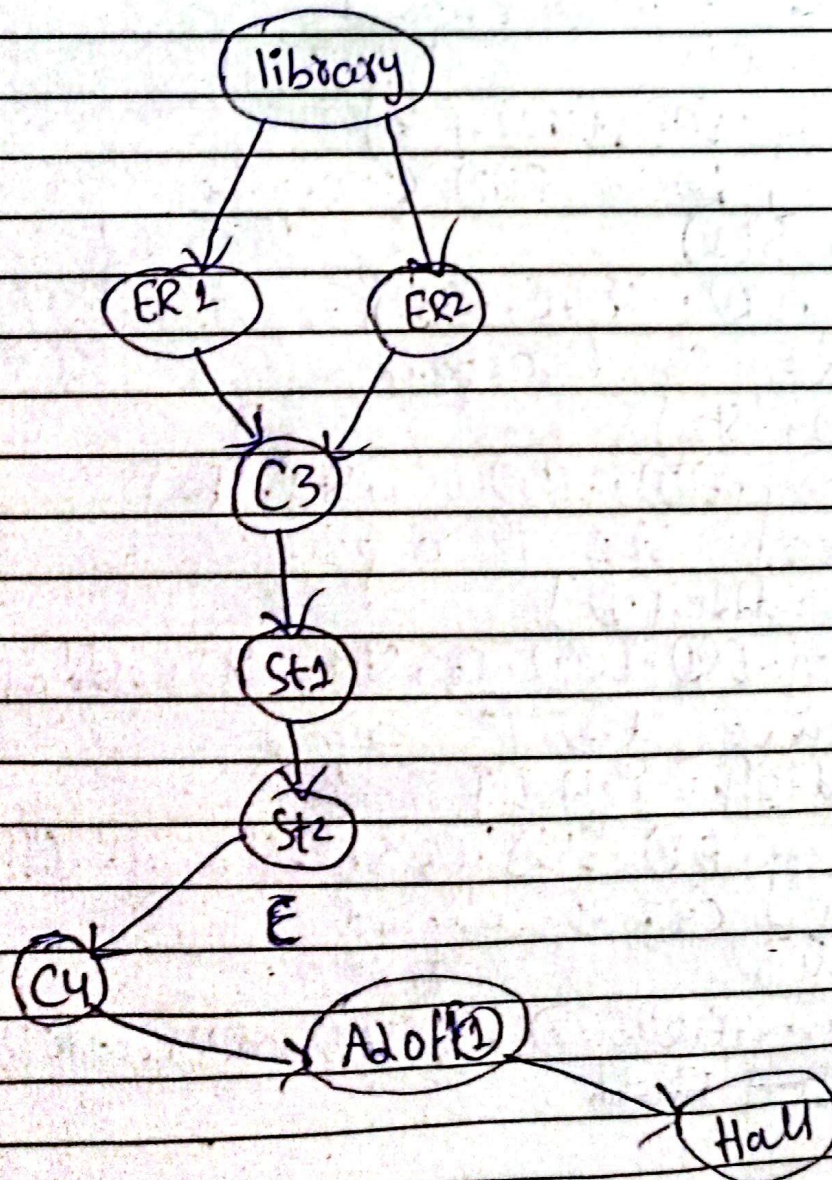
Initial goal $\Rightarrow$ library

goal state $\Rightarrow$ Hall

State Space $\Rightarrow$ Cs Dept

path Cost $\Rightarrow D=1$

Q find path from library to Hall



Find the shortest path using merge & Dequeue?

initializing:

Q $\Rightarrow$ [Library]

V $\Rightarrow$ [Library]

Dequeue Library & expand

Q $\Rightarrow$ [ER(1), ER(2)]

V $\Rightarrow$ [Library, ER(1), ER(2)]

Dequeue ER(1) & expand

Q $\Rightarrow$ [ER(2), C3]

V $\Rightarrow$ [Library, ER(1), ER(2), C3]

Dequeue ER(2) & expand

Q $\Rightarrow$ [C3, St1]

V $\Rightarrow$ [Library, ER(1), ER(2), C3, St1]

Dequeue C3 & expand

Q $\Rightarrow$ [St1, St2]

V $\Rightarrow$ [Library, ER(1), ER(2), C3, St1, St2]

Dequeue St1 & expand

Q $\Rightarrow$ [St2, Addoff(1)]

V $\Rightarrow$ [Library, ER(1), ER(2), C3, St1, St2, addoff(1)]

Dequeue St2 & expand

Q $\Rightarrow$ [addoff(1), Hall]

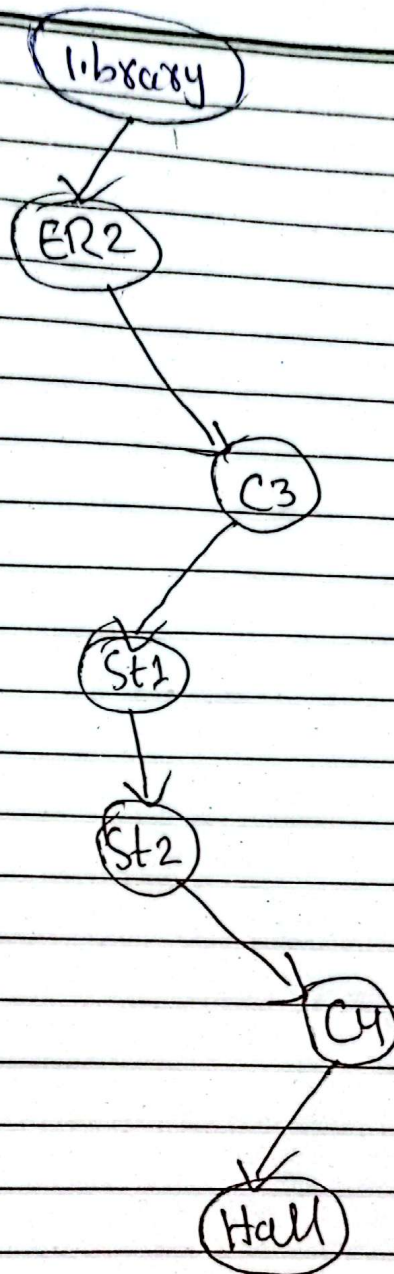
V $\Rightarrow$ [Library, ER(1), ER(2), C3, St1, St2, addoff(1), Hall]

Dequeue C4 & expand

Q $\Rightarrow$ [Hall]

V $\Rightarrow$ [Library, ER(1), ER(2), C3, St1, St2, addoff(1), C4, Hall]

Dequeue Hall



{ 7 nodes
6 edges }